Topic 15B: Reactions of transition metal elements
20. know the colours of the oxidation states of vanadium (+5, +4, +3 and +2) in its compounds
21. understand redox reactions for the interconversion of the oxidation states of vanadium (+5, +4, +3 and +2), in terms of the relevant $E^\ominus$ values
22. understand, in terms of the relevant $E^\ominus$ values, that the dichromate(VI) ion, $Cr_2O_7^{2-}$: i can be reduced to Cr^{3+} and Cr^{2+} ions using zinc in acidic conditions ii can be produced by the oxidation of Cr^{3+} ions using hydrogen peroxide in alkaline conditions (followed by acidification)
23. know that the dichromate(VI) ion, $Cr_2O_7^{2-}$, can be converted into chromate(VI) ions as a result of the equilibrium $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$
24. be able to record observations and write suitable equations for the reactions of Cr^{3+}(aq), Fe^{2+}(aq), Fe^{3+}(aq), Co^{2+}(aq) and Cu^{2+}(aq) with aqueous sodium hydroxide and aqueous ammonia, including in excess
25. be able to write ionic equations to show the difference between ligand exchange and amphoteric behaviour for the reactions in (24) above
26. understand that ligand exchange, and an accompanying colour change, occurs in the formation of: i $[Cu(NH_3)_4(H_2O)_2]^{2+}$ from $[Cu(H_2O)_6]^{2+}$ via $Cu(OH)_2(H_2O)_4$ ii $[CuCl_4]^{2-}$ from $[Cu(H_2O)_6]^{2+}$ iii $[CoCl_4]^{2-}$ from $[Co(H_2O)_6]^{2+}$
27. understand that the substitution of small, uncharged ligands (such as H_2O) by larger, charged ligands (such as Cl^-) can lead to a change in coordination number

Students should:
29. know that transition metals and their compounds can act as heterogeneous and homogeneous catalysts
30. know that a heterogeneous catalyst is in a different phase from the reactants and that the reaction occurs at the surface of the catalyst
31. understand, in terms of oxidation number, how V_2O_5 acts as a catalyst in the contact process
32. understand how a catalytic converter decreases carbon monoxide and nitrogen monoxide emissions from internal combustion engines by: i adsorption of CO and NO molecules onto the surface of the catalyst ii weakening of bonds and chemical reaction iii desorption of CO_2 and N_2 product molecules from the surface of the catalyst
33. know that a homogeneous catalyst is in the same phase as the reactants and appreciate that the catalysed reaction will proceed via an intermediate species
34. understand the role of Fe^{2+} ions in catalysing the reaction between I^- and $S_2O_8^{2-}$ ions
35. know the role of Mn^{2+} ions in autocatalysing the reaction between MnO_4^- and $C_2O_4^{2-}$ ions
CORE PRACTICAL 12: The preparation of a transition metal complex